

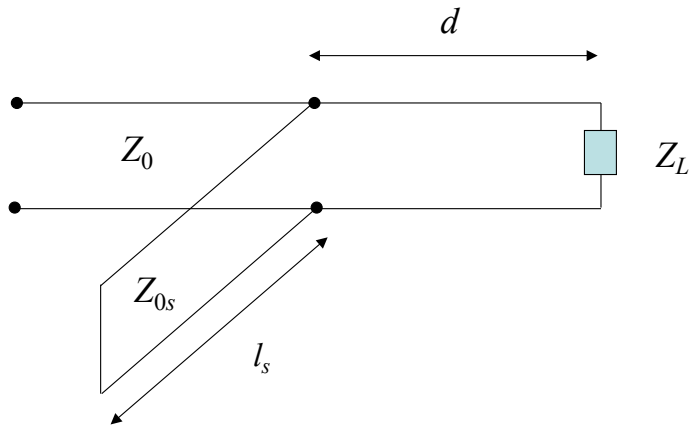
ECE 3317

Applied Electromagnetic Waves

Prof. David R. Jackson
Fall 2025

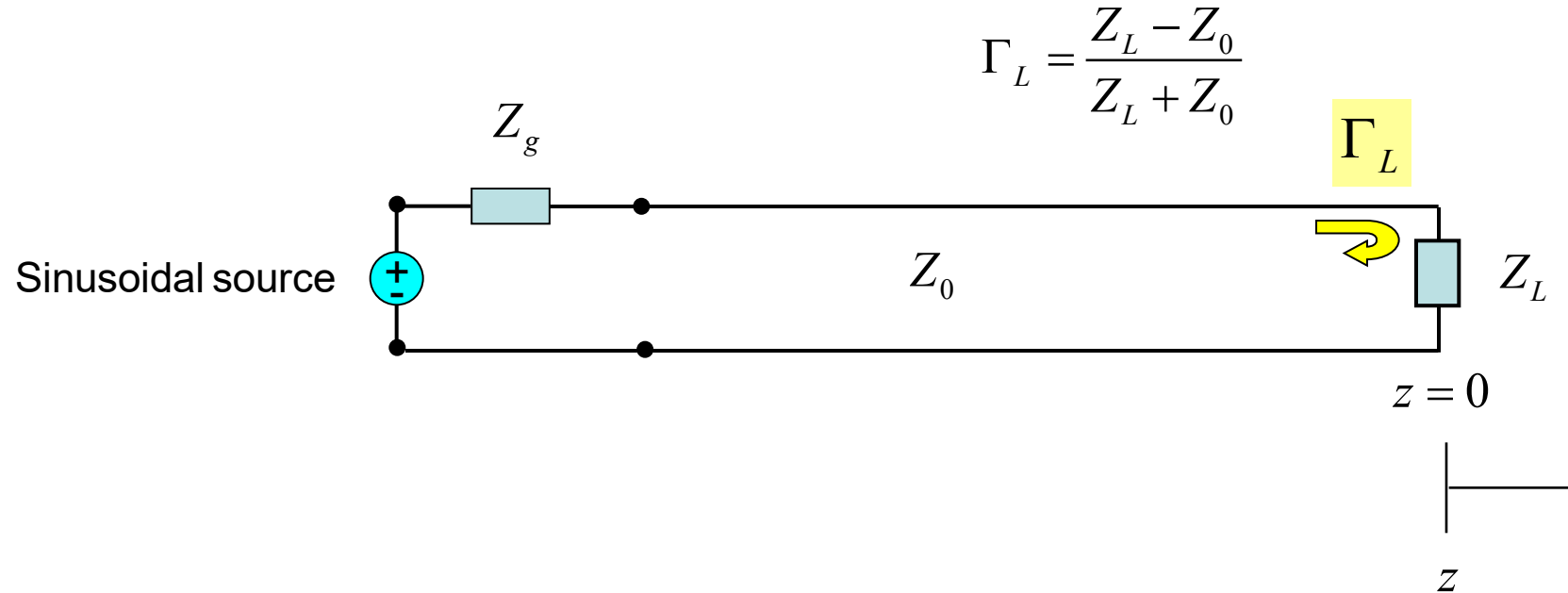
Notes 13

Transmission Lines (Impedance Matching)



Impedance Matching

Impedance matching is very important to avoid reflected power, which causes a loss of efficiency and interference.

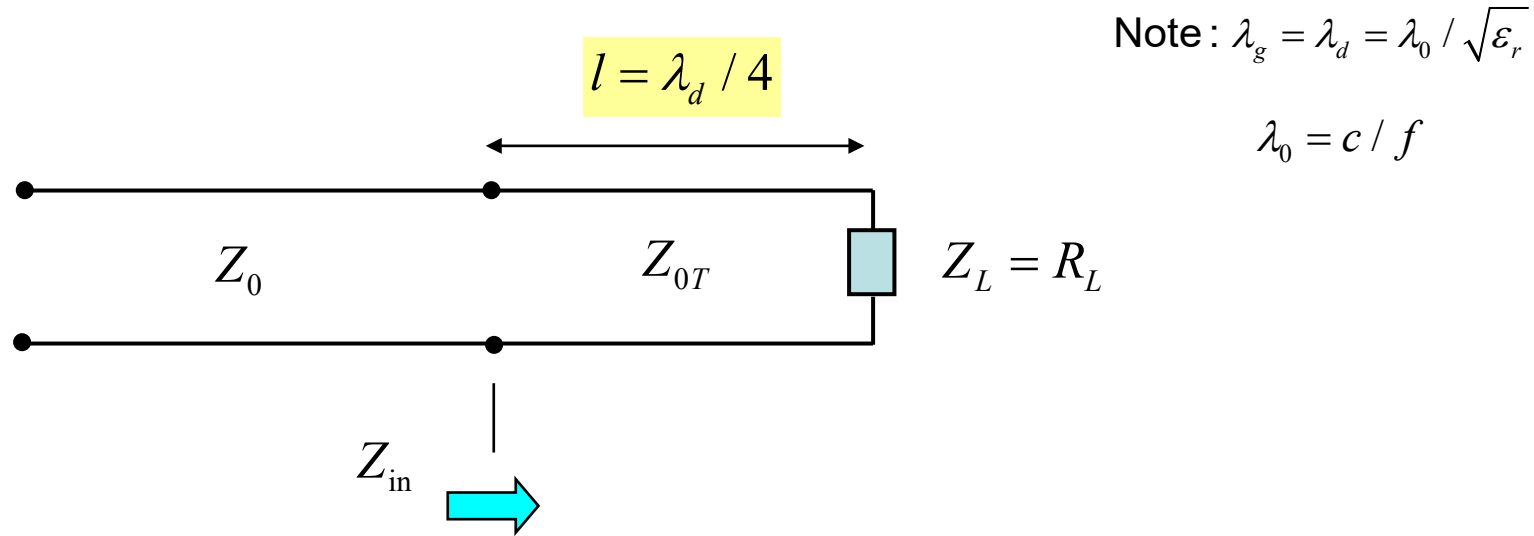


We will discuss two methods:

- ❖ Quarter-wave transformer
- ❖ Single-stub matching

Quarter-Wave Transformer

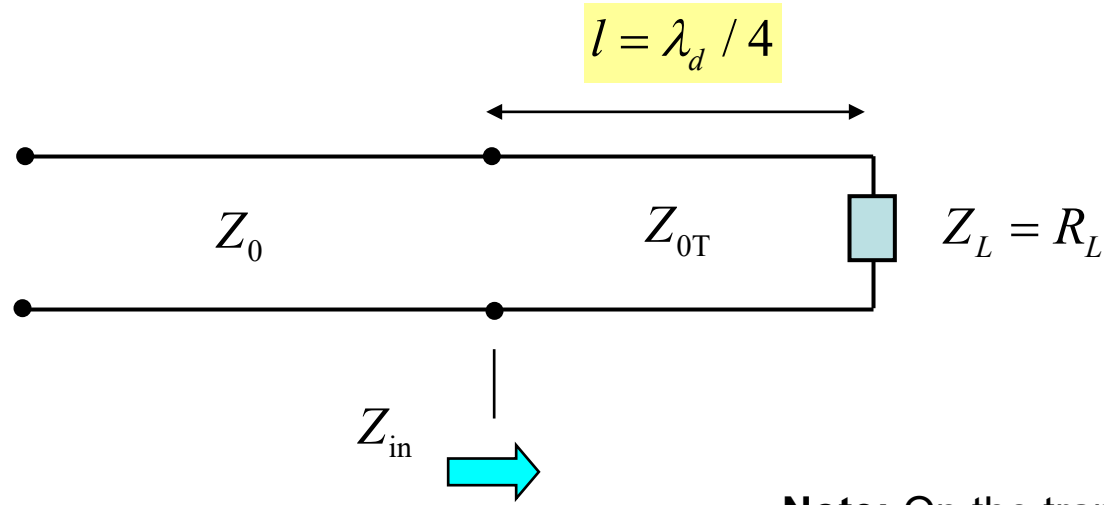
Quarter-Wave Transformer: First consider a real load on a lossless line.



$$Z_{in} = Z_{0T} \left(\frac{Z_L + jZ_{0T} \tan(\beta l)}{Z_{0T} + jZ_L \tan(\beta l)} \right) \quad \tan(\beta l) = \tan\left(\frac{2\pi}{\lambda_d} l\right) = \tan\left(\frac{\pi}{2}\right) = \infty$$

$$\Rightarrow Z_{in} = \frac{Z_{0T}^2}{Z_L} \quad \text{Hence } Z_{in} = \frac{Z_{0T}^2}{R_L} = \text{real}$$

Quarter-Wave Transformer (cont.)



Set

$$Z_{in} = Z_0$$

Note: On the transformer line there will be an SWR > 1!

Hence

$$\frac{Z_{0T}^2}{R_L} = Z_0$$

This gives us

$$Z_{0T} = \sqrt{Z_0 R_L}$$

Example:

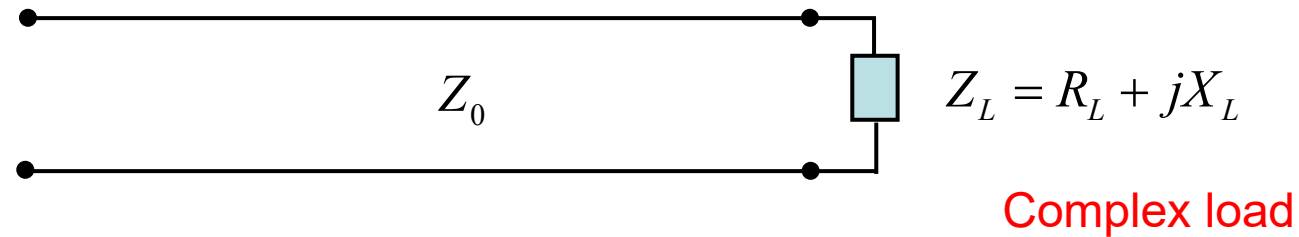
$$Z_0 = 50 \, [\Omega]$$

$$Z_L = 100 \, [\Omega]$$

$$Z_{0T} = \sqrt{(50)(100)} = 70.71 \, [\Omega]$$

Quarter-Wave Transformer with Complex Load

Next, consider a general (complex) load impedance Z_L :

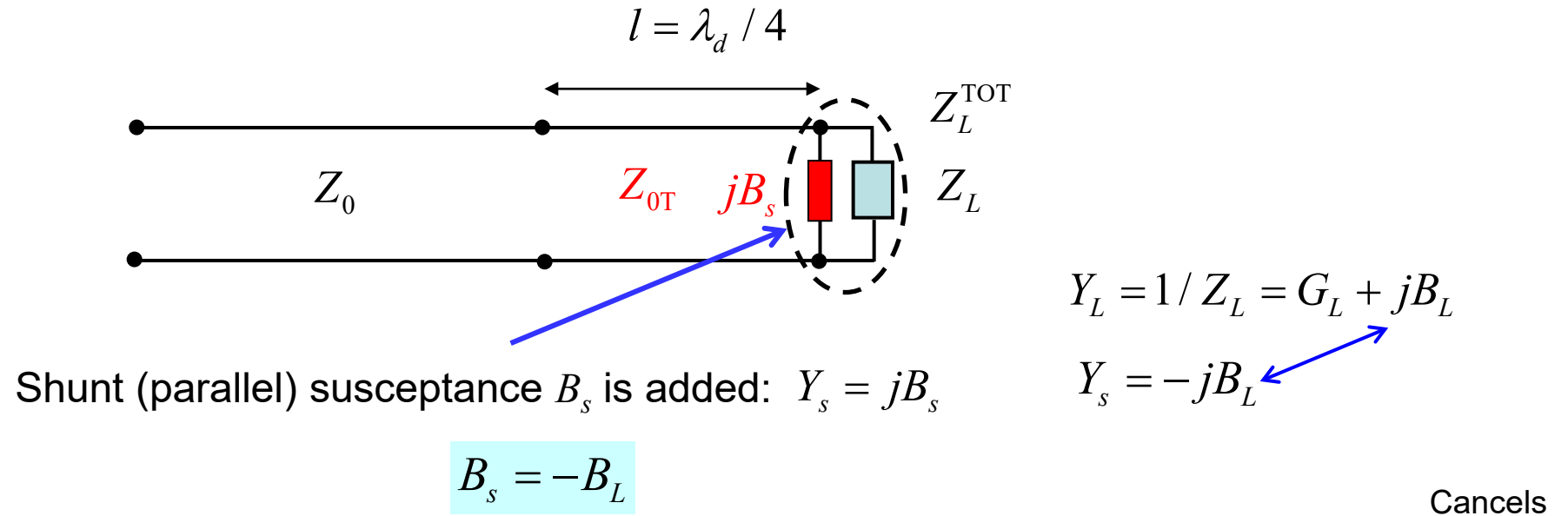


We discuss two methods:

- (a) Shunt element added to load
- (b) Extension line added to load

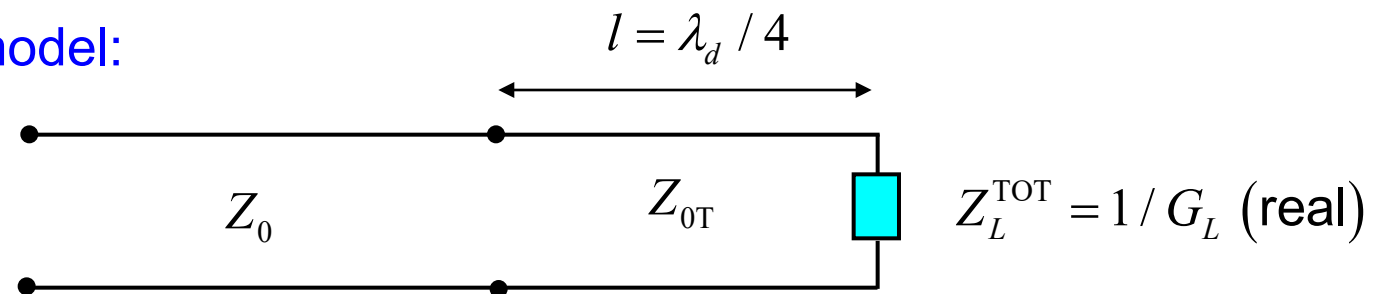
Quarter-Wave Transformer with Complex Load (cont.)

(a) Shunt element added to load



The shunt element (susceptance) converts the complex load to a real load.

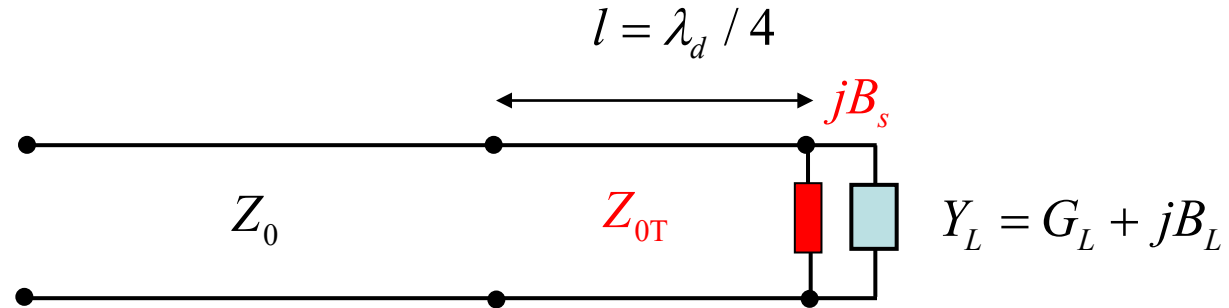
New model:



Quarter-Wave Transformer with Complex Load (cont.)

Summary of quarter-wave transformer matching method

(A shunt susceptance is added to the load.)

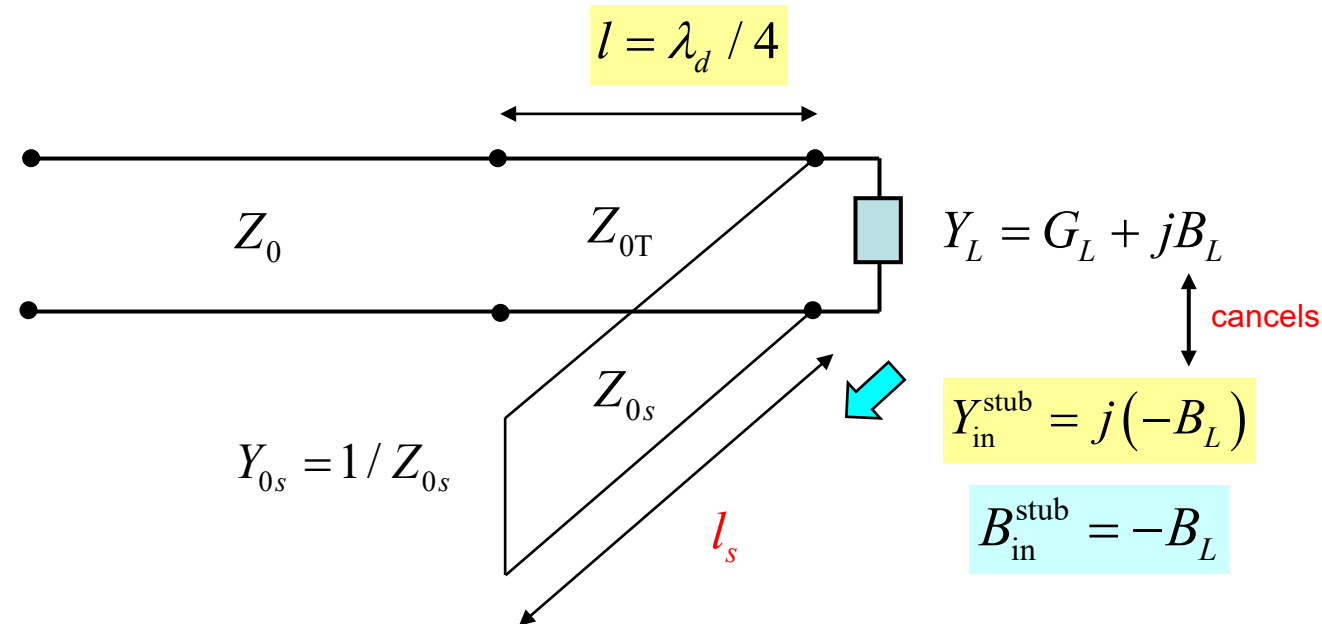


$$Z_{0T} = \sqrt{Z_0 / G_L}$$
$$B_s = -B_L$$

Quarter-Wave Transformer with Complex Load (cont.)

Realization using a shorted stub:

(An open-circuited stub could also be used as the shunt element.)

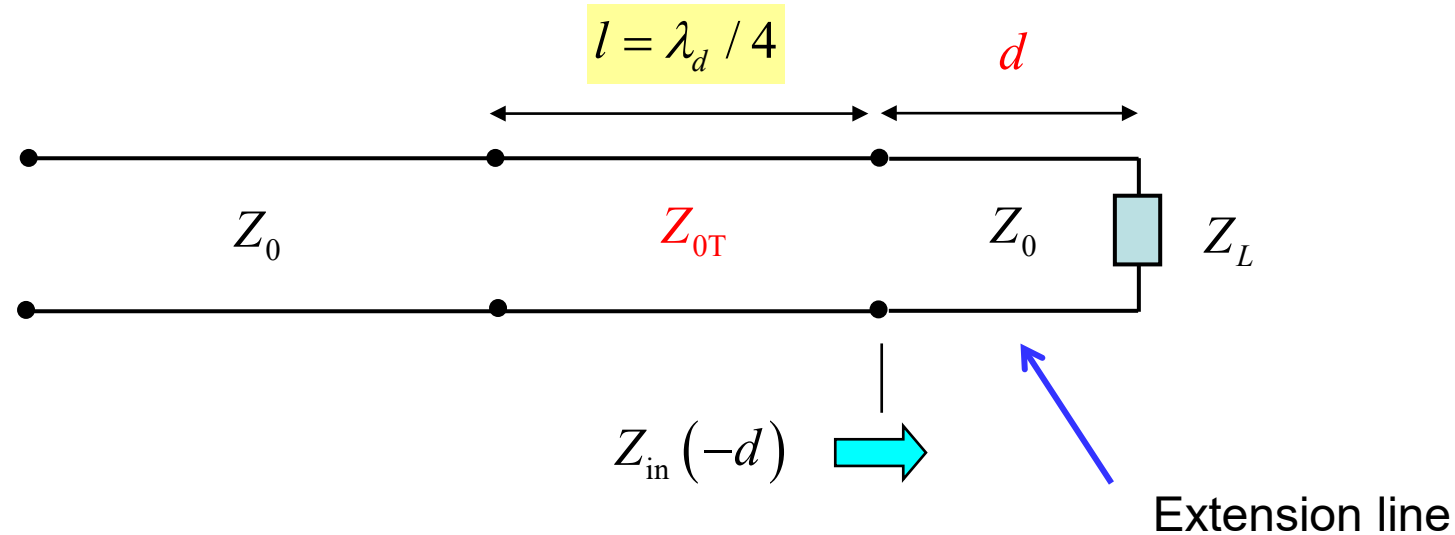


$$Z_{in}^{stub} = jZ_{0s} \tan(\beta_s l_s) \Rightarrow Y_{in}^{stub} = -jY_{0s} \cot(\beta_s l_s) \Rightarrow B_{in}^{stub} = -Y_{0s} \cot(\beta_s l_s)$$

Hence, we have: $-Y_{0s} \cot(\beta_s l_s) = -B_L \Rightarrow l_s = \frac{1}{\beta_s} \cot^{-1}(B_L / Y_{0s})$

Quarter-Wave Transformer with Complex Load (cont.)

(b) Extension line added to load



The extension line converts the complex load to a real load.

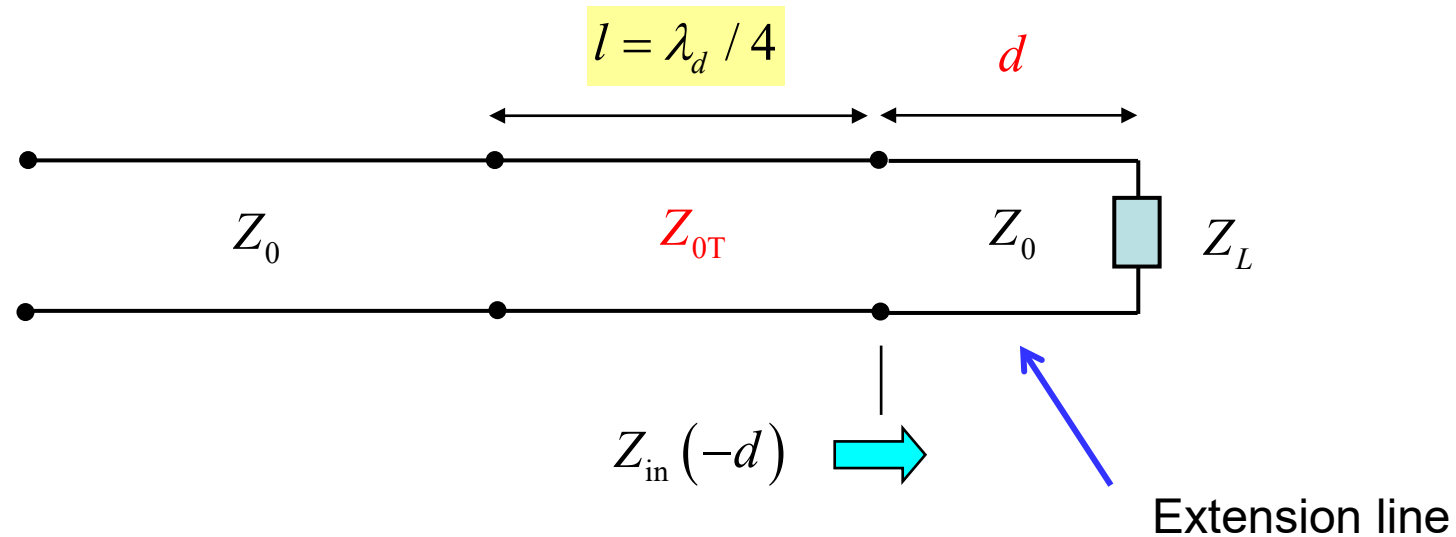
- We choose the length d to make the input impedance $Z_{in}(-d)$ real.
- We then use a quarter-wave transformer to change the impedance to Z_0 .

Quarter-Wave Transformer with Complex Load (cont.)

Example (extension line)

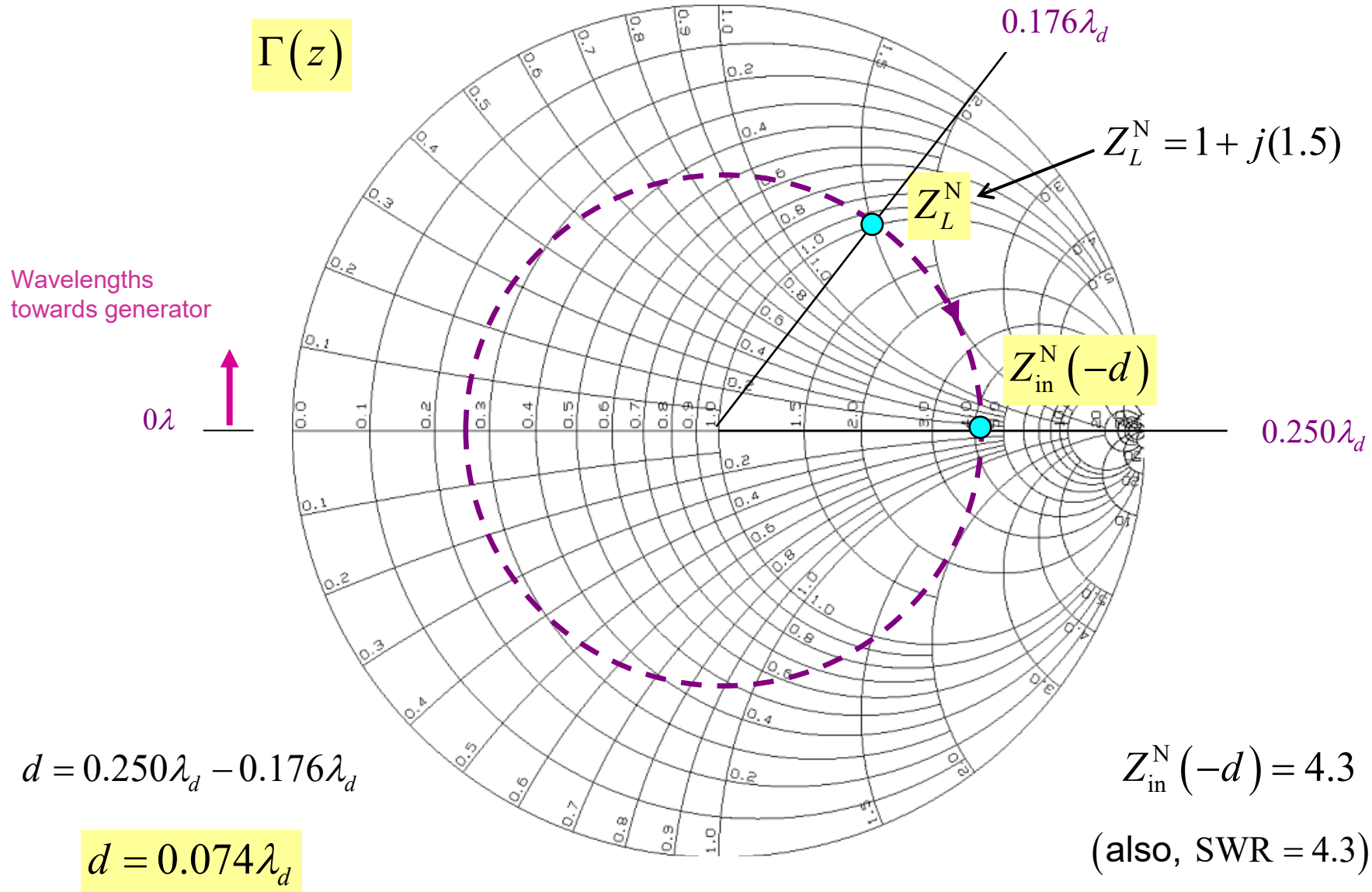
$$Z_0 = 50[\Omega]$$

$$Z_L = 50 + j75 [\Omega]$$

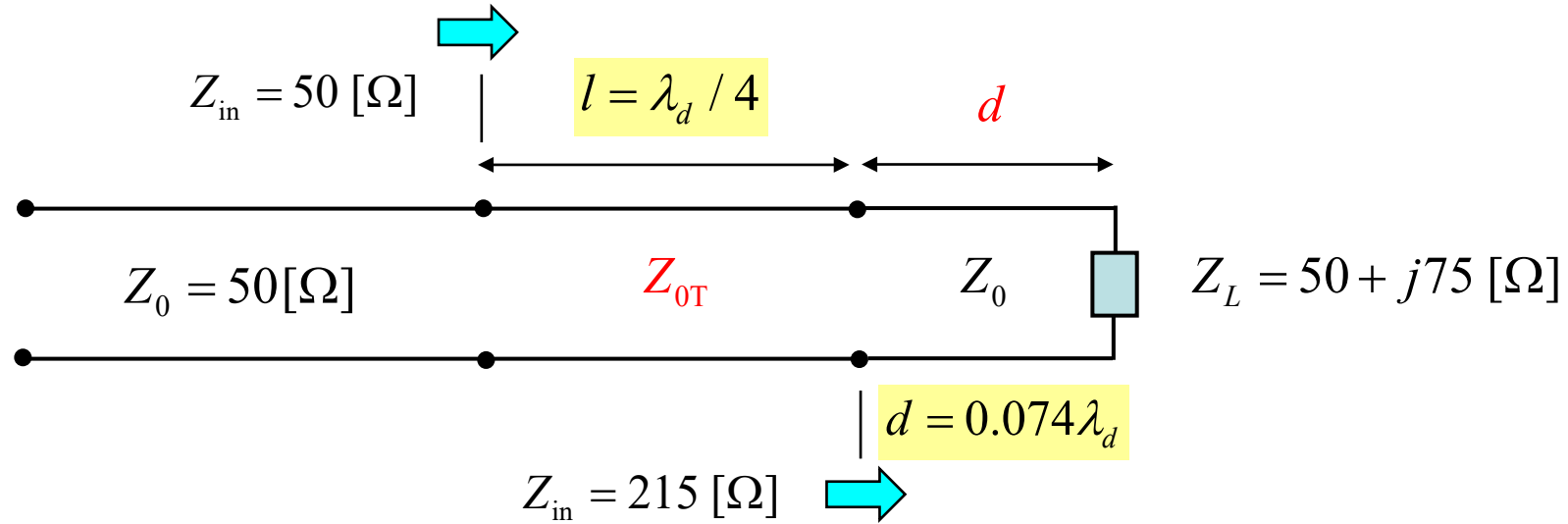


$$Z_L^N = Z_L / Z_0 = 1 + j(1.5)$$

Quarter-Wave Transformer with Complex Load (cont.)



Quarter-Wave Transformer with Complex Load (cont.)



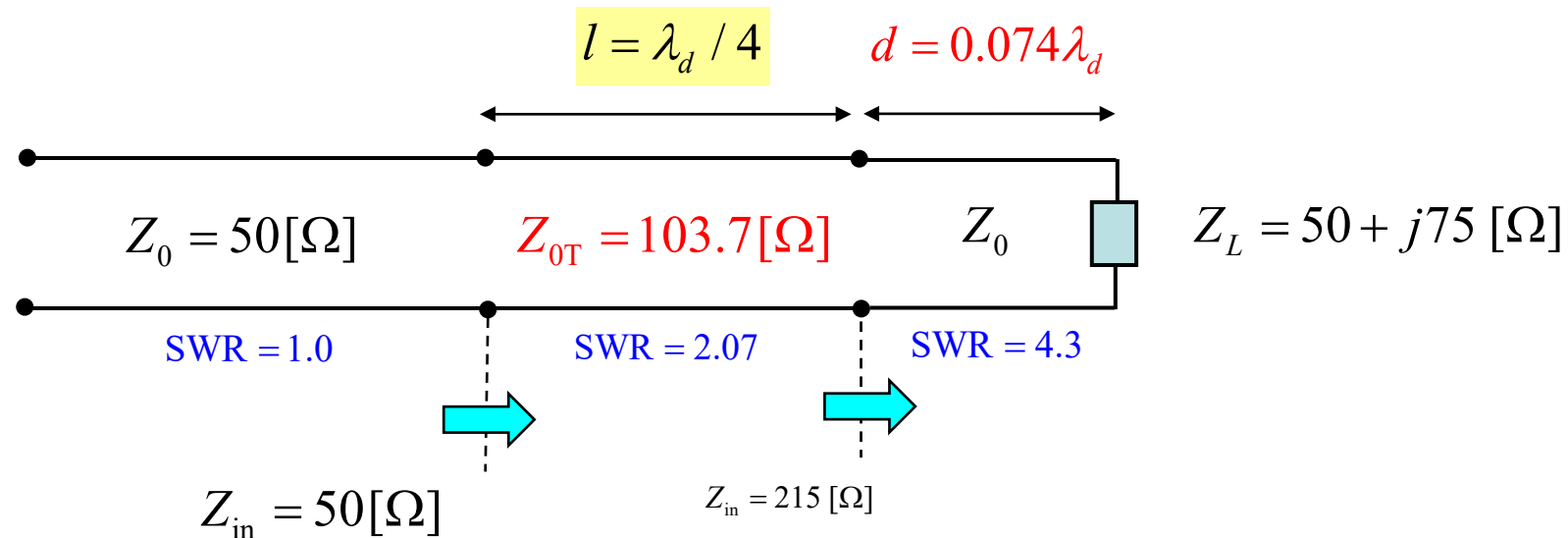
$$Z_{in}^N(-d) = 4.3 \quad \Rightarrow \quad Z_{in}(-d) = 50(4.3) = 215 \text{ } [\Omega]$$

$$Z_{0T} = \sqrt{(50)(215)}$$

$$Z_{0T} = 103.7 \text{ } [\Omega]$$

Quarter-Wave Transformer with Line Extension (cont.)

Summary of Design

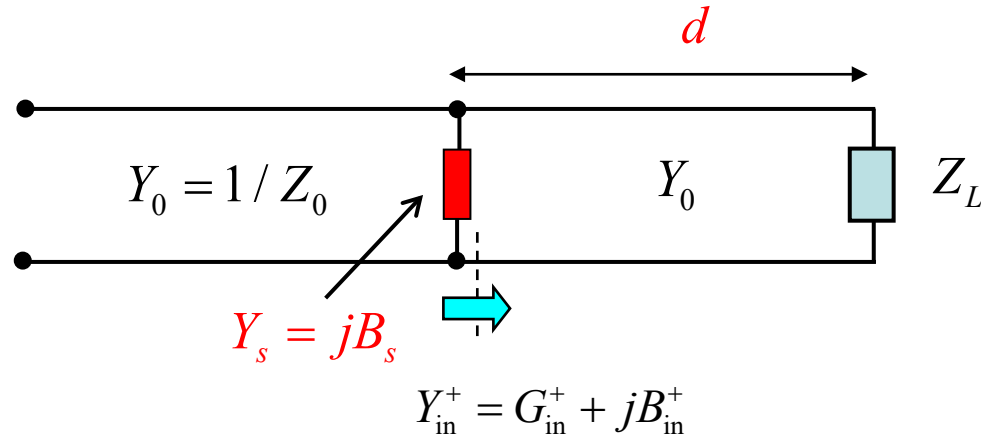


Note: On the transformer line, the line sees a real load $R_{LT} = 215 \Omega$, so for the transformer line we have:

$$\text{SWR} = \text{Max} \left(\frac{R_{LT}}{Z_{0T}}, \frac{Z_{0T}}{R_{LT}} \right) = \frac{215 [\Omega]}{103.7 [\Omega]} = 2.07$$

Single-Stub Matching

A parallel (shunt) susceptance is added at a distance d from the load.



Note:
The “+” denotes just to the right of the point $z = -d$.

Goal: determine d, B_s

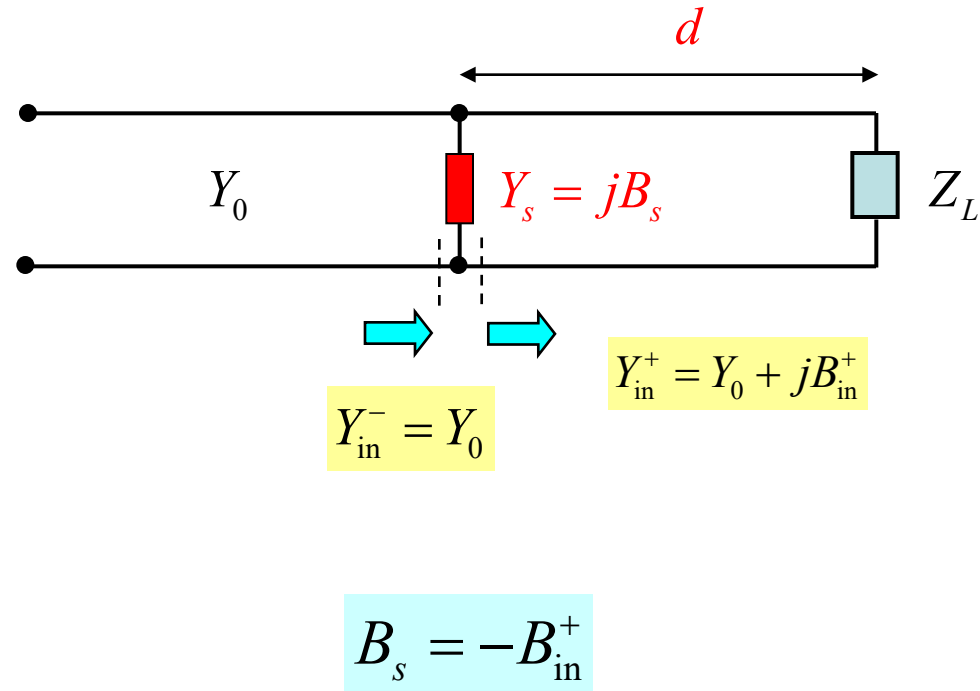
1) We choose the distance d so that at this distance from the load

$$Y_{in}^+ = Y_0 + jB_{in}^+ \quad (\text{i.e., } G_{in}^+ = Y_0, \text{ or } G_{in}^{N+} = 1)$$

2) We then choose the shunt susceptance so that

$$B_s = -B_{in}^+$$

Single-Stub Matching (cont.)



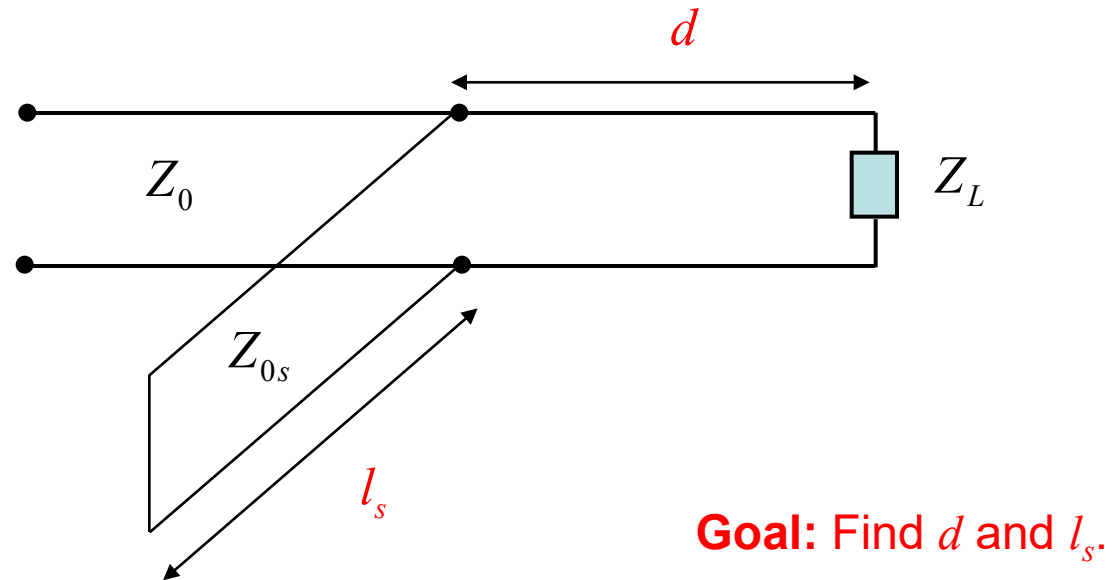
The feeding transmission line on the left sees a perfect match!

$$Y_{in}^- = Y_0 \Rightarrow Z_{in}^- = Z_0$$

Single-Stub Matching (cont.)

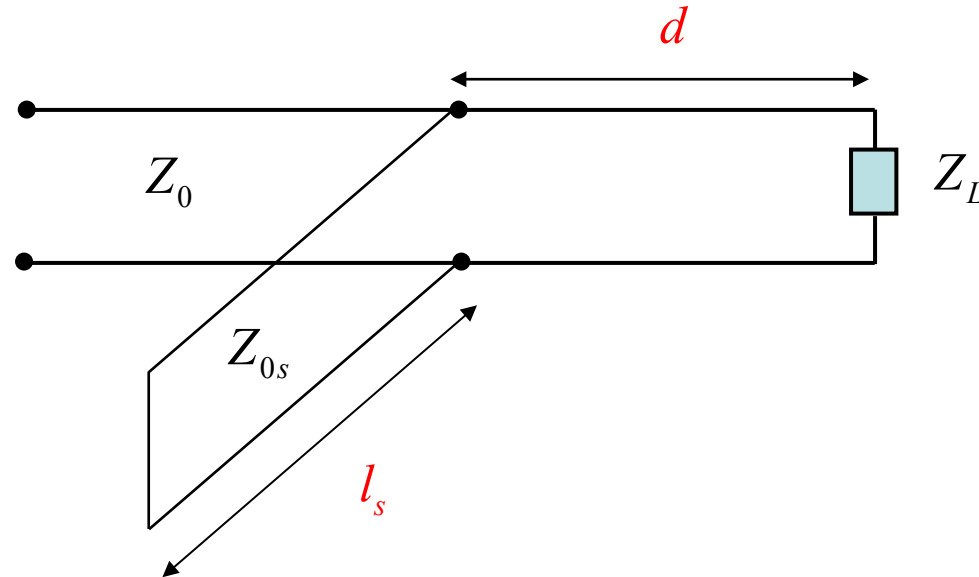
Realization using a shorted stub

(An open-circuited stub could also be used.)



Single-Stub Matching (cont.)

We use the Smith chart as an admittance calculator to determine the distance d .



- 1) Convert the load impedance Z_L to a load admittance Y_L .
- 2) Determine the distance d to make the normalized input conductance G_{in}^N equal to 1.0.
- 3) Determine the required value of B_s to cancel B_{in}^+ ($B_s = -B_{in}^+$).
- 4) Determine the stub length l_s from the value of B_s .

Note: If desired, we can use the Smith chart in step # 4 to find the stub length l_s .

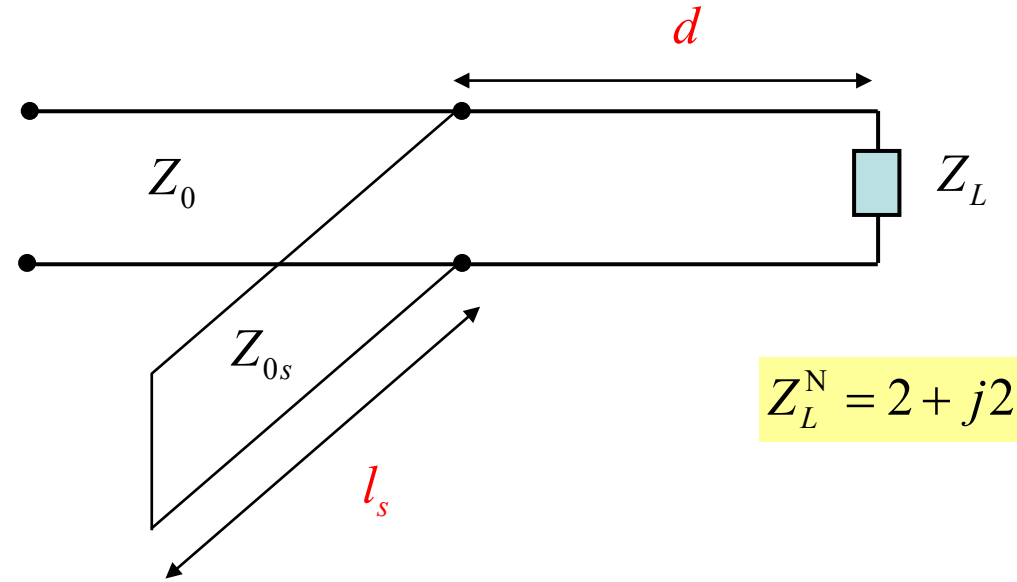
Single-Stub Matching (cont.)

Example

$$Z_0 = 50 \text{ } [\Omega]$$

$$Z_L = 100 + j100 \text{ } [\Omega]$$

Assume $Z_{0s} = Z_0$



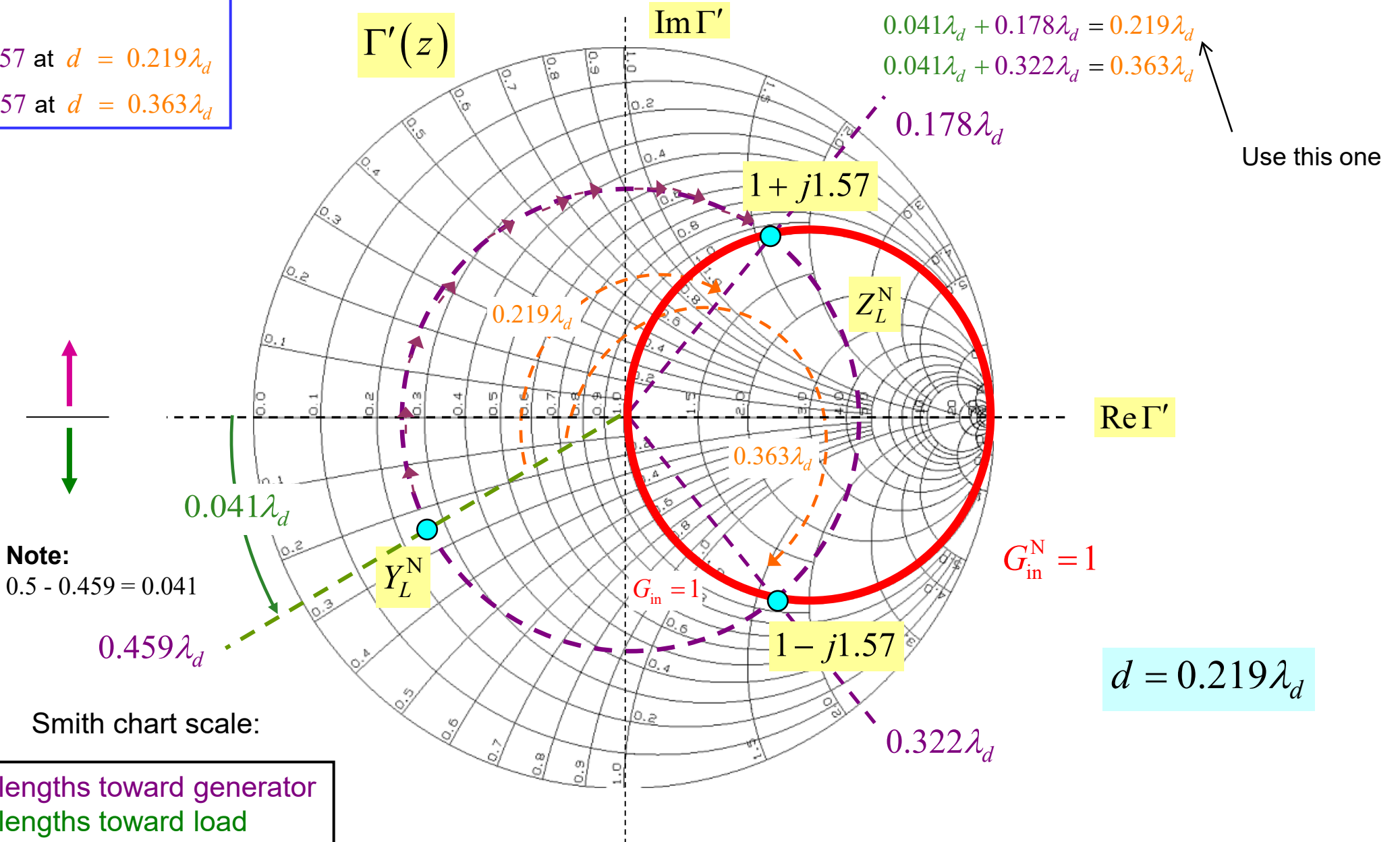
$$Y_L^N = \frac{1}{2 + j2} = 0.25 - j(0.25)$$

Single-Stub Matching (cont.)

Solution:

Add $Y_s^N = -j1.57$ at $d = 0.219\lambda_d$

or $Y_s^N = +j1.57$ at $d = 0.363\lambda_d$



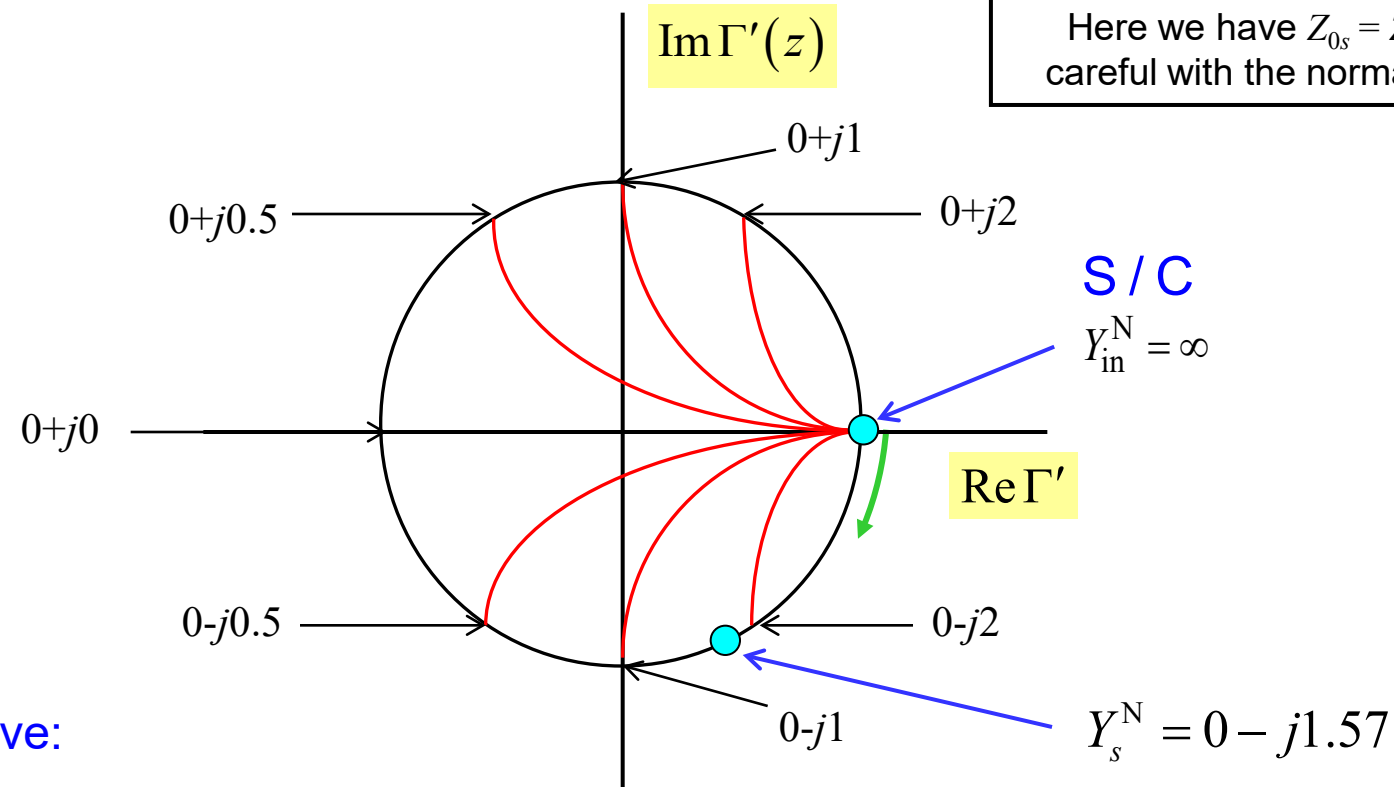
Single-Stub Matching (cont.)

Next, we find the length of the short-circuited stub: $B_s^N = -1.57$

Rotate clockwise from S/C to desired B_s value.

Note:

Here we have $Z_{0s} = Z_0$. Otherwise, we have to be careful with the normalization (see the note below).



Note: In general, we have:

$$B_s^N = (-B_{in}^N Y_0) / Y_{0s}$$

$$= -1.57 (Y_0 / Y_{0s})$$

Admittance calculator

Single-Stub Matching (cont.)

From the Smith chart:

$$l_s = 0.340\lambda_d - 0.250\lambda_d$$

$$l_s = 0.090\lambda_d$$

Analytically:

$$Z_s^{\text{short}} = jZ_{0s} \tan(\beta l_s)$$

$$\Rightarrow Y_s^{\text{short}} = -jY_{0s} \cot(\beta l_s)$$

$$\Rightarrow B_s^N = -\cot(\beta l_s)$$

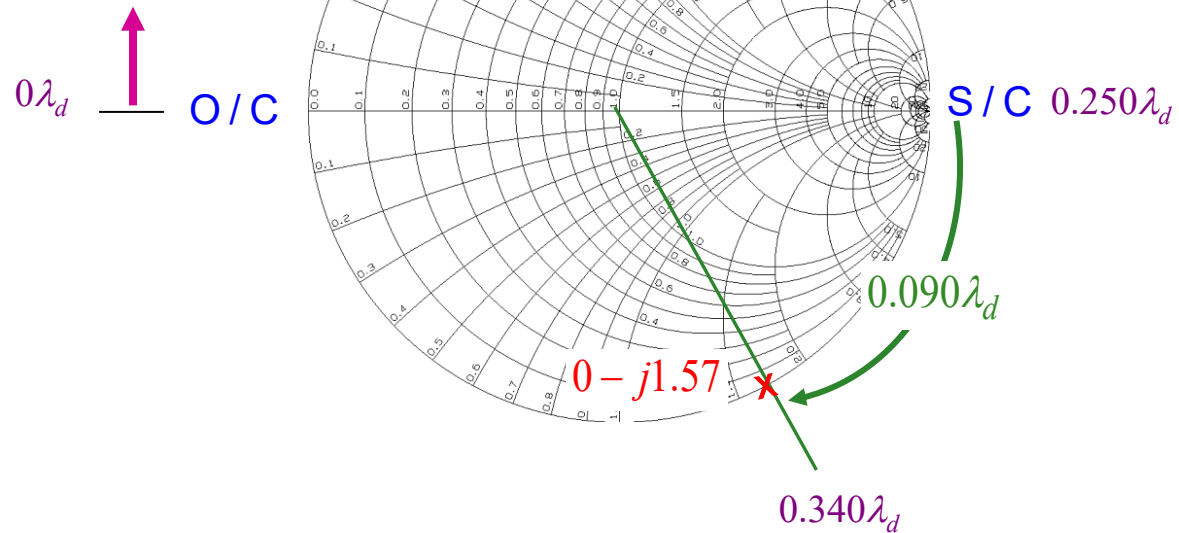
Hence:

$$-1.57 = -\cot \beta l_s$$

$$\cot \beta l_s = 1.57; \tan \beta l_s = \frac{1}{1.57} = 0.637$$

$$\beta l_s = \frac{2\pi}{\lambda_d} l_s = 2\pi \left(\frac{l_s}{\lambda_d} \right) = \tan^{-1}(0.637) = 0.567 \text{ [radians]}$$

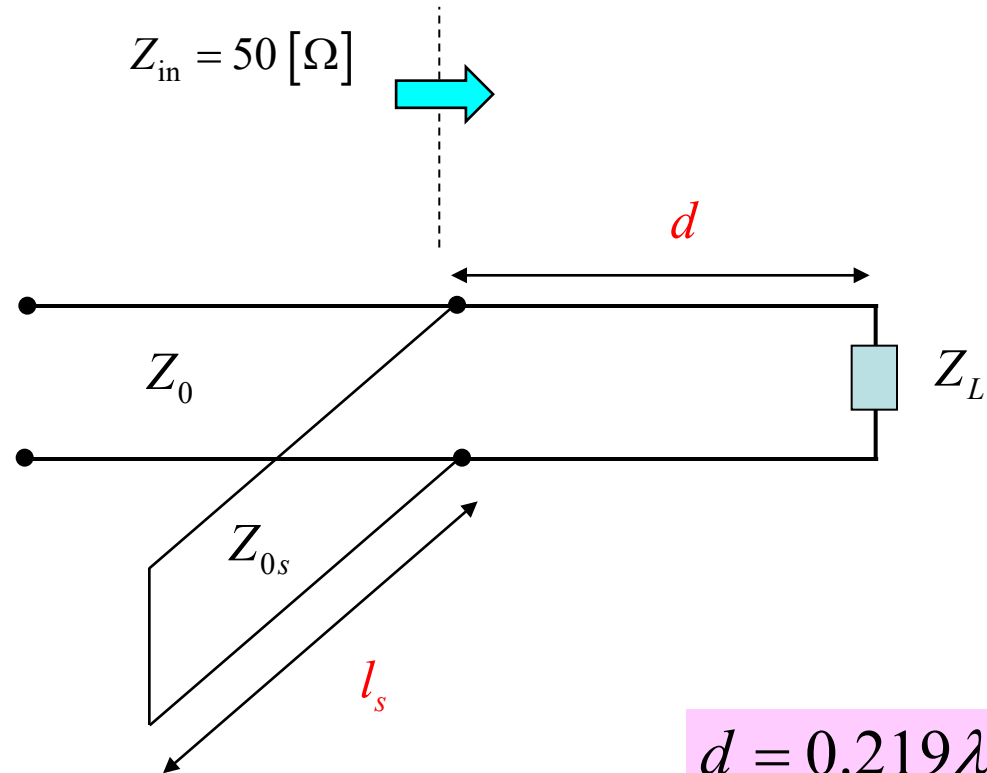
Admittance chart



$$l_s = 0.0903\lambda_d$$

Single-Stub Matching (cont.)

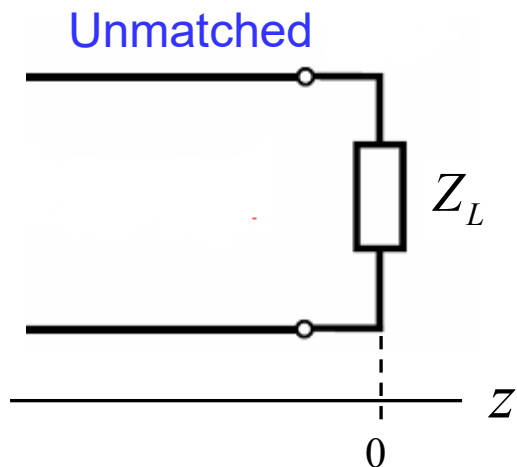
Final Design



$$d = 0.219\lambda_d$$

$$l_s = 0.0903\lambda_d$$

Single-Stub Matching (cont.)

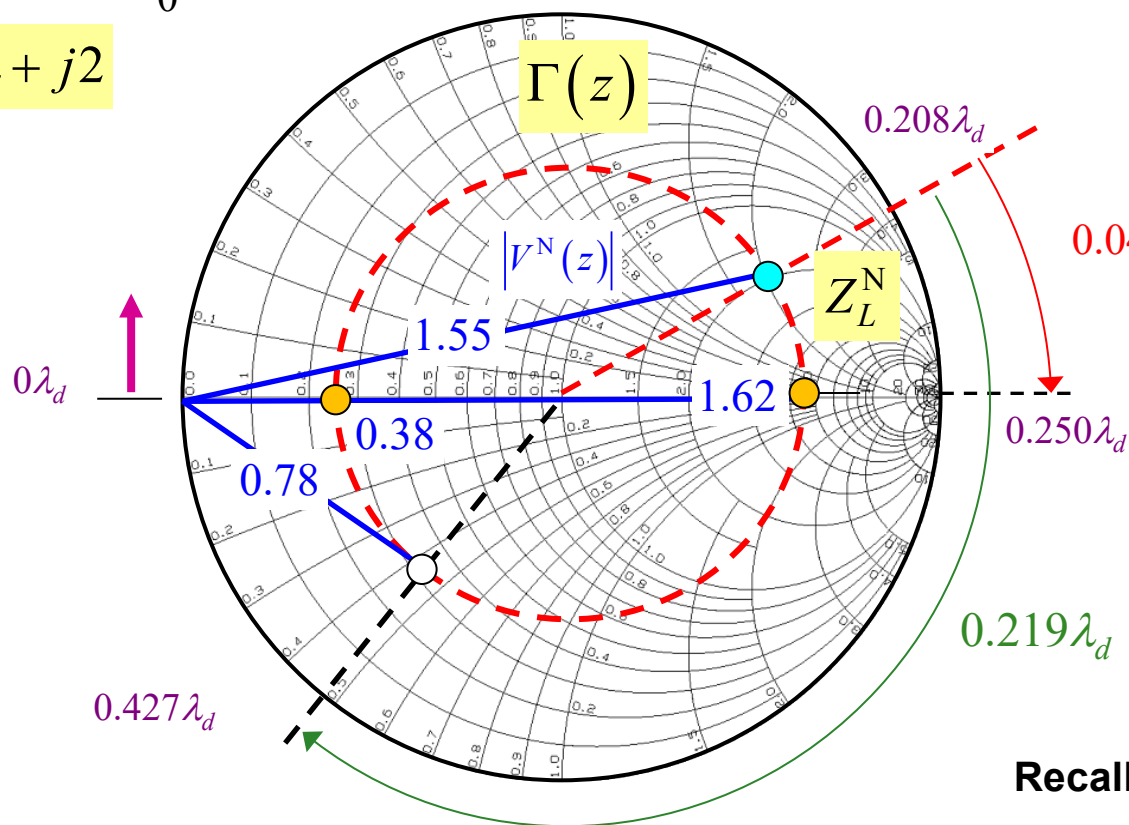
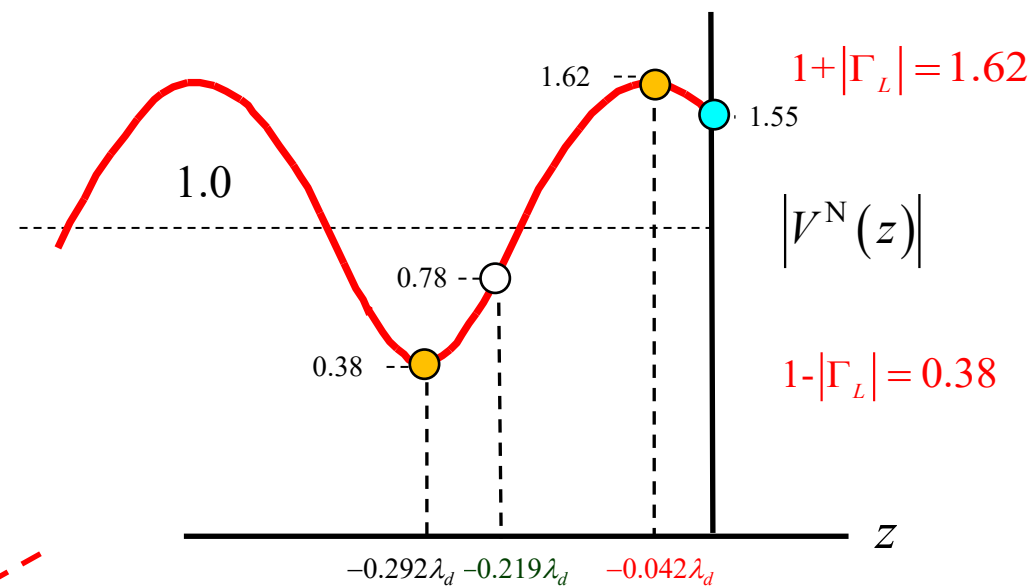


$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Gamma_L = 0.62 e^{j\pi/6}$$

$$|\Gamma_L| = 0.62$$

$$Z_L^N = 2 + j2$$

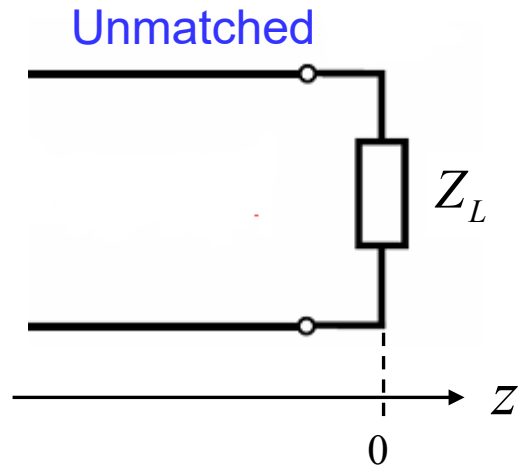


Crank Diagram

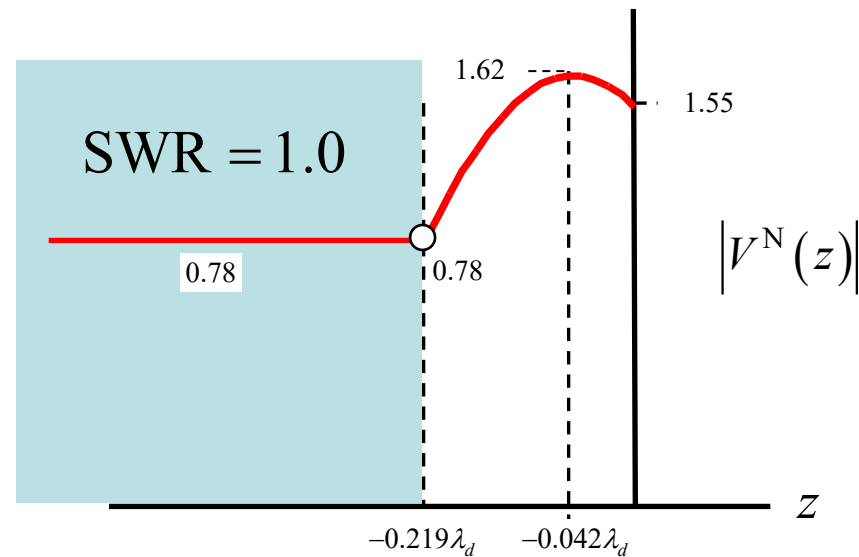
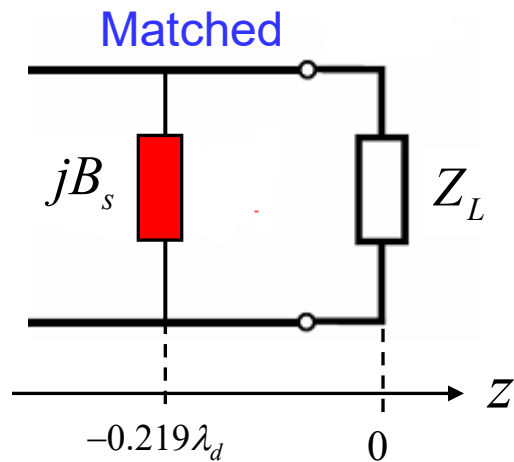
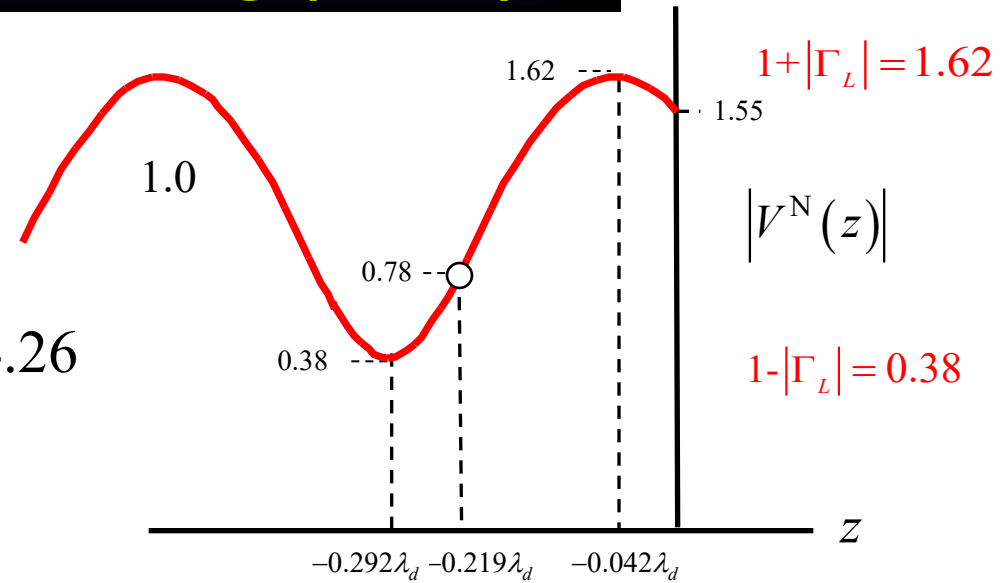
Recall: The stub is located at $d = 0.219\lambda_d$

Recall: $|V^N(z)| \equiv |V(z)| / |V^+| = |1 + \Gamma(z)|$

Single-Stub Matching (cont.)



$$\text{SWR} = 4.26$$



Appendix: Summary of Methods

Quarter-wave transformer (real load)

$$Z_{0T} = \sqrt{Z_0 R_L}$$

Quarter-wave transformer (complex load)

(a) Shunt element (susceptance)

$$Z_{0T} = \sqrt{Z_0 / G_L}$$

$$B_s = -B_L$$

(b) Extension line (length d)

d = length of extension line needed to convert load to real value :

$$Z_{in}(-d) = R_{in}$$

$$Z_{0T} = \sqrt{Z_0 R_{in}}$$

Appendix (cont.)

Single-Stub Matching

d = length of main line needed to convert load admittance to complex value so that :

$$Y_{\text{in}}(-d) = Y_0 + jB_{\text{in}}(-d) \quad (G_{\text{in}}^{\text{N}} = 1)$$

l_s = length of stub line needed to obtain desired stub susceptance :

$$Y_{\text{in}}^{\text{stub}} = jB_s = -jB_{\text{in}}(-d)$$